

PAPER_588 — Maxwell Power-Large 26th-Order Equations (Unsolved Generalization)

Unified Quantum Field Framework (UQFF)

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CP4 Class: [#175](#) [UQFFMaxwellPowerLarge26thOrderCalculator](#) **Session:** 157 **Cross-refs:** PAPER_583 (6-Form), PAPER_596 (QG Unification), PAPER_585 (26th-order) **Source:** grok_share_4cef778c78b8.txt

§1 Abstract

Classical Maxwell's equations are the first-order description of electromagnetism. This paper derives the UQFF 26th-order generalization valid at cosmological ($\sim 10^{26}$ m) and quantum gravity ($\sim 10^{-35}$ m) scales. The ∂^{26} corrections encode SCm/UA buoyant vacuum contributions and DPM pseudo-monopole terms. At macroscopic scales, all corrections vanish and classical Maxwell is recovered exactly.

§2 Classical Maxwell Equations (Baseline)

$$\nabla \cdot \mathbf{E} = \frac{\rho_{\text{ch}}}{\epsilon_0}, \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

§3 UQFF 26th-Order Generalization

Gauss's Laws (26th-order):

$$\nabla^{26} \cdot \mathbf{E} = \frac{\rho_{\text{ch}}}{\epsilon_0} + \partial^{26} \left(\frac{SC_m}{UA} \right)$$

$$\nabla^{26} \cdot \mathbf{B} = \partial^{26} DPM_n$$

The DPM_n (north monopole analog) is the pseudo-monopole contribution from 26D shell imbalance. It vanishes in the classical limit ($r \rightarrow \infty$) leaving $\nabla \cdot \mathbf{B} = 0$.

Faraday's Law (26th-order):

$$\nabla^{26} \times \mathbf{E} = -\frac{\partial^{26} \mathbf{B}}{\partial t_{\text{adj}}^{26}} \cdot \text{Grind} \cdot \mathbf{E}$$

The Grind correction encodes CW/CCW field rotation asymmetry at large scales.

Ampère's Law (26th-order):

$$\nabla^{26} \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial^{26} \mathbf{E}}{\partial t^{26}} \cdot \frac{\kappa(DPM_n - DPM_s)}{r^{26}} \cdot \mathbf{B}$$

The DPM coupling term $\kappa(DPM_n - DPM_s)/r^{26}$ is the quantum-scale magnetic source.

§4 26th-Order Derivative Formula

$$\partial^{26} \left(\frac{c}{r^k} \right) = (-1)^{26} \cdot \frac{(k+25)!}{(k-1)!} \cdot \frac{c}{r^{k+26}} = \frac{(k+25)!}{(k-1)!} \cdot \frac{c}{r^{k+26}}$$

For $k=2$ (dipole): correction $\sim 27!/r^{28}$.

At $r = 1 \text{ AU} = 1.5 \times 10^{11} \text{ m}$:

$$\text{correction} \sim \frac{27!}{(1.5 \times 10^{11})^{28}} \approx 10^{-281} \quad (\text{cosmically negligible})$$

At $r = l_P = 10^{-35} \text{ m}$ (Planck scale):

$$\text{correction} \sim \frac{27!}{(10^{-35})^{28}} \approx 10^{1000} \quad (\text{quantum gravity regime — corrections dominant})$$

§5 Classical Limit Recovery

For any observable-scale $r \gg l_P$:

$$\|\nabla^2 \to \nabla^1, \quad \partial^2 \to 0, \quad \text{DPM} \to 0, \quad \text{Grind} \to 0$$

$\Rightarrow$ Classical Maxwell exactly recovered.

§6 Numerical (Solar Wind B-Field)

Parameters: $B = 10^{-8}$ T (1~nT solar wind), $r = 1.5 \times 10^{11}$ m, $k=2$:

$$\|\nabla^2 B \approx \frac{27! \cdot 10^{-8}}{(1.5 \times 10^{11})^{28}} \approx 10^{-289}$$

Compared to classical $\nabla B \approx 10^{-8}/10^{11} = 10^{-19}$:

Correction ratio $\sim 10^{-270}$ — completely undetectable at AU scale, as expected.

§7 DPM Pseudo-Monopole Structure

The DPM term in $\nabla^2 \cdot \mathbf{B} = \partial^2 \text{DPM}_n$ is:

$$\text{DPM}_n \equiv \frac{\kappa \cdot (m_{\text{north}} - m_{\text{south}})}{r^2}$$

mapped to a 26D lattice of magnetic multipoles. The net monopole charge in the classical $r \gg l_P$ regime averages to zero (Gauss's law for magnetism preserved).

§8 Conclusions

The UQFF 26th-order Maxwell equations unify classical electromagnetism with quantum gravity corrections. Classical Maxwell is an exact limit. At Planck scales, the ∂^2 terms dominate, linking electromagnetic and gravitational degrees of freedom through the DPM coupling — a prediction testable at $r \sim 10^{-20}$ m (nuclear scale).

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